WRICNet: A Weighted Rich-Scale Inception Coder Network for Remote Sensing Image Change Detection

Yu Jiang, Lei Hu, *Member, IEEE*, Yongmei Zhang, and Xin Yang

***Abstract*— The majority of remote sensing image change detection models focus on extracting high-level semantic features. However, it is difficult to detect the changing area with large differences in size simultaneously, and the accurate changing area edge is difficult. To solve these problems, we propose a weighted rich-scale inception coder network (WRICNet), consisting of the weighted rich-scale inception module and the weighted rich-scale coder module. The former can retain the low-level multiscale feature (LMF), and the latter can extract the high-level multiscale feature (HMF). By fusing LMF and HMF, both large and small changing areas can be detected and make changing area edge accurate. To complement LMF and HMF well, we propose a weighted scale block, which assigns appropriate weights to multiscale features. Compared to comparative methods, performance experiments on datasets demonstrate that our proposed method can further reduce false alarms and miss alarms and make the changing area edge more accurate. Furthermore, ablation studies show that our training strategy, model settings, and improvements of the inception and rich-scale blocks are effective.**

***Index Terms*— Change detection (CD), convolutional neural network (CNN), deep learning (DL), multiscale, remote sensing (RS).**

I. INTRODUCTION

CHANGE detection (CD) can be defined as the identification of differences in the state of an object or a pattern observed at different times [1]. With the development of Earth observation technology, an increasing number of satellite sensors (e.g., IKONOS, GaoFen (GF), SPOT, and QuickBird) have been designed to acquire remote sensing (RS) images. Therefore, it is becoming increasingly popular to analyze change information through the advanced RS CD technology. RS CD can be applied in diverse applications, such as urbanization [2], [3], forest monitoring [4], [5], water [6], shifting land use [7], and disasters [8], [9].

RS CD methods evolved with the development of machine learning. In the early stage, to detect changes of different sizes or shapes, a method based on support vector machines (SVMs) [10] uses the local wavelet coefficients of the image block as the basic feature and integrates the SVM-based classifier to CD. Compared with SVM, extreme learning machine (ELM) has better learning ability and generalization ability. To reduce the loss of useful information in the difference image generation process, a method uses ELM [11] train images patch centered at pixels that are generated by neighborhood-based ratio. To effectively and automatically discriminate between changed and unchanged pixels in the difference image, a method based on Markov random field (MRF) [12] analyzes the difference image by considering the spatial-contextual information included in the neighborhood of each pixel. Compared to MRF, which relies on the local feature of the image, a conditional random field (CRF) can extract the global feature of the image. To improve robustness against noise and preserve boundary information, a method based on CRF [13] uses the memberships of unsupervised fuzzy C-means clustering algorithm, and pairwise potential adopts a boundary constraint based on the Euclidean distance. Limited by priors, the generalization ability of the abovementioned traditional RS CD methods is insufficient, and it is difficult to meet the current accuracy requirements.

With the development of deep learning, the convolutional neural network (CNN) has become an increasingly popular RS CD method. Limited by the size of the convolution kernel, the receptive field (RF) of the feature is small. Multiscale is particularly important for extracting a large RF, and it is effective in various fields [14]–[16]. In addition, due to the difference in the size of the changing area in RS images, it is particularly important to represent the features of different scales. There are multiple methods for extracting multiscale features. For instance, InceptionNets [17]–[20] uses multiple groups, each of which has different RFs, to extract features of different scales. FPN [14] is a top-down architecture with lateral connections and exploits the inherent multiscale, pyramidal hierarchy of deep convolutional networks to extract feature pyramids. Based on FPN, PANet [21] uses the bottom-up path augmentation to enhance the ability to extract low-level features. UNet [22] also uses a top-down architecture to extract multiscale features, and unlike FPN, UNet uses concatenate to fuse features at the same level, which can reduce the loss of features. Res2Net [15] uses the rich-scale block construct hierarchical residual-like connections within a single residual

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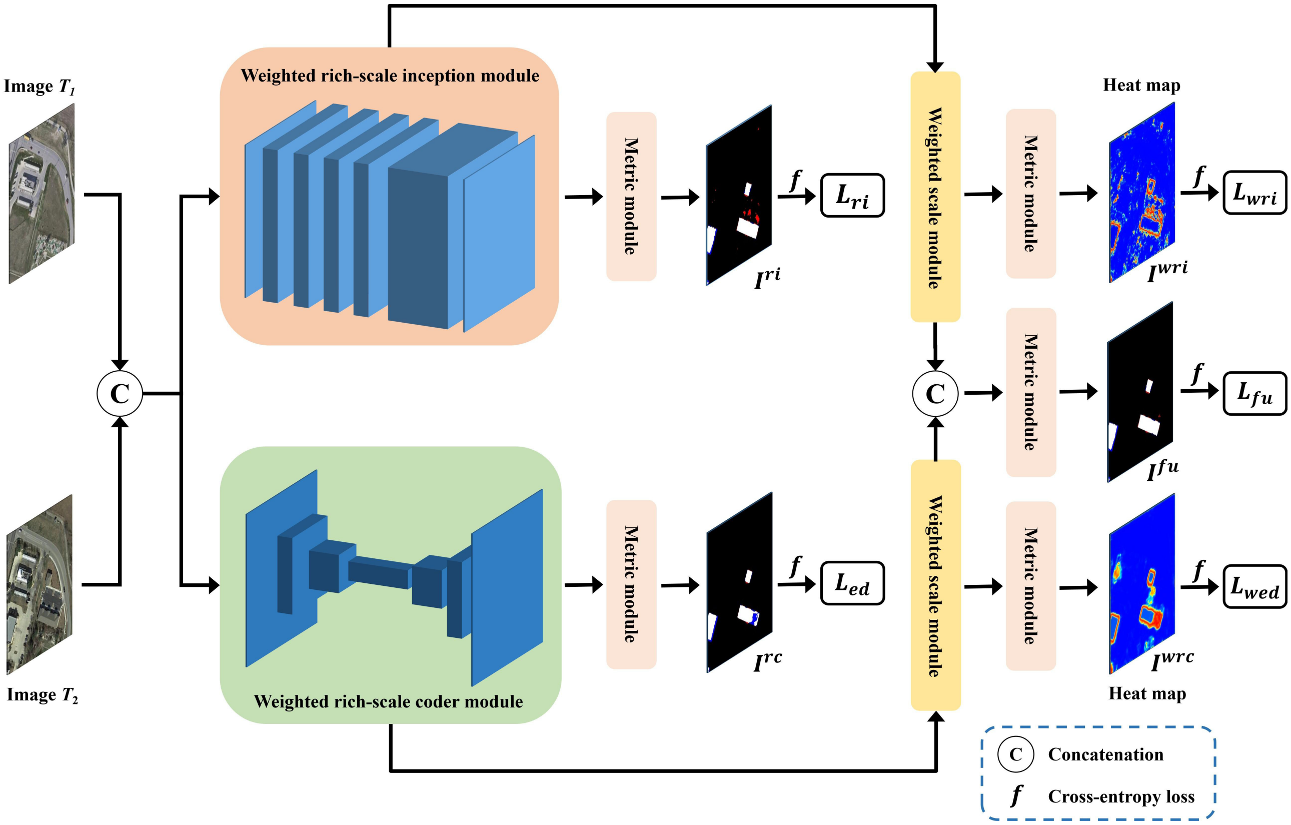

Fig. 1. Overview of WRICNet. Image T_1 and Image T_2 as the input, concatenating channels first and then use the WRI module and WRC module to extract LMFs and HMFs, respectively. After assigning adaptive weights to multiscale features via weighted scale module, concatenating channels of weighted multiscale features, and obtaining the binary change result I^{fu} via metric module, in addition, we also obtain I^{ri}, I^{rc}, I^{wri}, and I^{wrc} to calculate the jointed cross-entropy loss. In I^{ri}, I^{rc}, and I^{fu}, black indicates a correctly detected unchanged pixel, white indicates a correctly detected change, red indicates a change detected in error, and blue indicates an unpredicted change. To analyze the weight distribution, we obtain the heat map of I^{wri} and I^{wrc}, red indicates a large weight, and blue indicates a small weight.

block, which makes obtaining multiscale features feasible, and UNet++ [23] uses the nest structure to fit the optimal depth and redesigns the skip connection for extracting multiscale features.

Due to the lack of satisfactory features that can effectively distinguish the changing area, the features are usually sensitive to factors such as noise, angle, shadow, and context, so the method has low robustness to pseudo changes [24]. The attention mechanism (AM) has been proven to be effective in solving this problem [24], [25]. For instance, the self-attention mechanism (Self-AM) [26] explores the performances of nonlocal operations of images and videos in the space–time dimension, which is of substantial significance for the long-range dependencies of images. The spatial attention mechanism (SAM) [27] can enhance feature representations by encoding the context information of a long range into local features. In addition, the channel attention mechanism (CAM) [28] can establish the relationships of features between channels.

In RS CD, the low-level feature is beneficial to retain high-frequency features (e.g., small change area, especially changing area edge). Meanwhile, there is more noise, which will result in fewer miss alarms (MAs) and more accurate edges of the changing area, but more false alarms (FAs). The high-level feature is beneficial for retaining low-frequency features (semantic information) and can filter noises, but will lose part of high-frequency features, which is reflected in fewer FAs, and the edge of the changing area is not accurate, but more MAs.

The above methods can extract high-level features by multiple downsample (DS) and upsample (US) modules, which will cause low-level features lost. Although there are methods to enhance the fusion of high- and low-level features between DS and UP modules [21], [23], that is still insufficient. Recently, EGRCNN [29] obtained the changing area edge of the ground truth via the canny operator and added the edge to the input of the deep neural network to make up for the low-level feature lost. In addition, the above method does not consider the deterioration of the results after the fusion when fusing multiscale, low- and high-level features. For example, after fusing low- and high-level features, FAs and MAs may be retained (but we expect to reduce FAs and MAs).

To solve the above two problems, we design an end-to-end network (WRICNet). First, we directly use two branches to extract the low-level multiscale feature (LMF) and high-level multiscale feature (HMF). Second, we improve multiscale feature extraction blocks and design a weighted scale module to assign appropriate weights to multiscale, low- and high-level features. The main contributions of this article are summarized as follows.

1) We use two branches to extract the LMF and HMF. The LMF can retain more details (e.g., the changing area edge and small changing area). The HMF can extract more semantic information. Complementing each other can further reduce FAs and MAs.
2) We improve the module for extracting multiscale features. We reduce the number of parameters of the inception module and enhance the ability of rich-scale block to express the channelwise feature.
3) We propose a weighted scale block, which can assign adaptive pixelwise and channelwise weights to features and benefit for fusing multiscale, low- and high-level features correctly.

The rest of this article is organized as follows. The proposed methodology is presented in Section II. The datasets, experimental settings, experimental performance, and comparison against the comparative methods are discussed in Section III. Finally, we conclude in Section V.

II. METHODOLOGY

A. *Weighted Rich-Scale Inception Coder Network*

As shown in Fig. 1, our proposed WRICNet takes images T_1 and T_2 as input and performs concatenation between images T_1 and T_2 in the channel dimension before feature extraction. WRICNet consists of four components: a weighted rich-scale inception (WRI) module, a weighted rich-scale coder (WRC) module, a weighted scale block, and a metric module.

The WRI module and the WRC module can obtain multiscale features. The former does not change the size of the feature in the process of obtaining the multiscale feature, and the input feature and the output feature have the same dimension. Therefore, we call the obtained feature LMF, which has the characteristics of fewer MAs but more FAs. The latter first reduces the feature size and then enlarges the feature size to obtain multiscale features. Furthermore, the input feature and the output feature have the same dimension. Thus, we call the obtained feature HMF, which has the characteristics of more MAs but fewer FAs. The weighted scale block assigns appropriate weights to LMF and HMF and then fuses them via concatenation. Finally, the metric module with a densely connected structure [30] is used to obtain the CD output.

B. *Weighted Scale Block*

There are different meanings in different positions and channel dimensions since the multiscale feature is a set of local features. For instance, LMF has more FAs, while HMF has fewer. The changing area of LMF is complete, while HMF has many MAs. The changing area edge of LMF is more accurate than that of HMF. We expect to reduce the weight of LMF outside the changing area and increase that of HMF to reduce FAs after feature fusion. Therefore, it is extremely important to obtain the weight of the multiscale feature in the position and channel dimensions. As shown in Fig. 2, we design the weighted scale block to assign appropriate pixelwise and channelwise weights to the multiscale feature. x is the input, $w(x)$ is the pixelwise and channelwise weight of feature x, and $f(x)$ is used to obtain the attention map of multiscale features. We use normalization, Conv block-1, and activation to get $w(x)$ since $w(x)$ is related to the channel so that we use "LayerNormation" for normalization. Conv block-1 can calculate the weight of feature x. We set the initial value of the weight to 1. Therefore, in the initial state, $f(x) = x$. We use "sigmoid" as the activation function to calculate $w(x)$ and squeeze the weight to [0, 1] to keep $f(x)$ normalized.

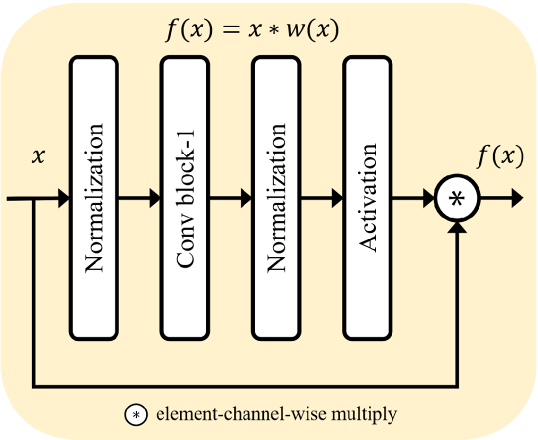

Fig. 2. Weighted scale block.

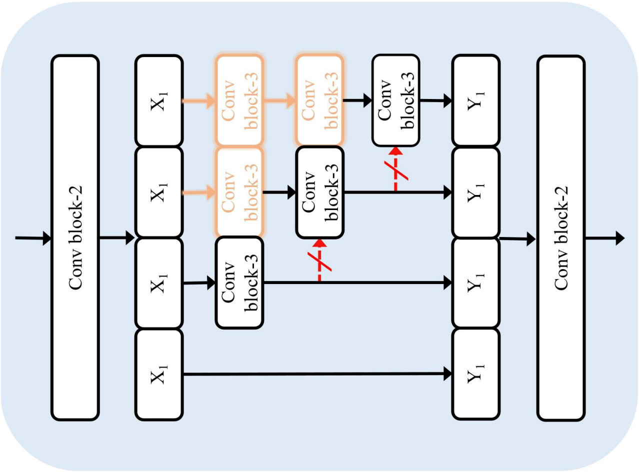

Fig. 3. Rich-scale block. The yellow part is added by us and the red part is deleted by us.

C. *Rich-Scale Block*

To strengthen the ability of the model to obtain multiscale features, Gao *et al.* [15] designed a rich-scale block, which can be plugged into state-of-the-art backbone CNNs. The rich-scale block divides the input feature into multiple groups in the channel dimension and extracts the features of different scales in each group. Therefore, the feature extracted by the rich-scale block is channelwise multiscale. The latter group first fuses the features obtained by the previous group and then performs feature extraction so that each group can obtain

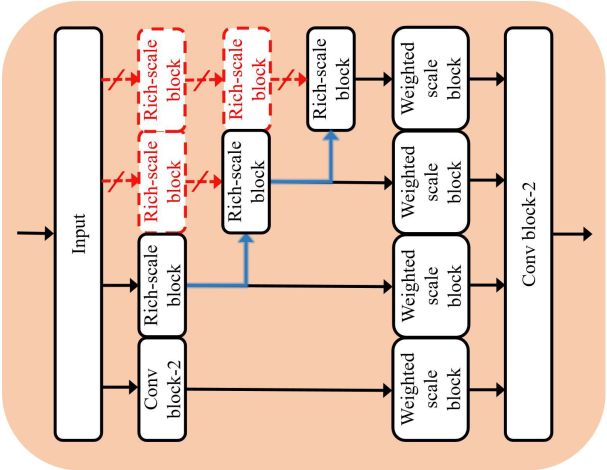
Input
Rich-scale block
Rich-scale block
Rich-scale block
Rich-scale block
Rich-scale block
Rich-scale block
Conv block-2
Weighted scale block
Weighted scale block
Weighted scale block
Weighted scale block
Conv block-2

Fig. 4. WRI module. The blue line is added by us and the red part is deleted by us.

different RFs via different numbers of convolutions to obtain multiscale features.

Since the input feature of each group is different, without changing the scale obtained by each group, as shown in Fig. 3, we use different numbers of Conv block-3 to obtain multiscale features in each group rather than fusing the features obtained by the previous group, which can strengthen the independence of each group feature and the expression ability of multiscale features. According to Gao *et al.* [15], we set the scale (group) of rich-scale block to 4. The kernel_size, stride, padding, and activation of Conv block-3 are set to 3, 1, “same,” and “relu,” respectively.

D. WRI Module

Based on the Hebbian theory, which introduces sparse features and extracts multiscale features by widening the network structure, Szegedy *et al.* [17], Ioffe and Szegedy [18], and Szegedy *et al.* [19], [20] designed the inception module. In this structure, each branch uses a different number of convolutions to obtain the features of different scales [31].

Since the input of each branch is the same, the feature scales obtained after the same number of convolutions on each branch are the same. According to this prior knowledge, we regard the extracted features of each branch, except for the first branch, as the input of the next branch. Therefore, the model parameters are reduced by reducing the number of convolutions in the branch.

As shown in Fig. 4, to further improve the ability to extract multiscale features, we design the WRI module, which replaces the convolution block in the inception module with the rich-scale block. The weighted scale block assigns appropriate weights to the features extracted by each branch and then fuse features of different scales via concatenation. In the first branch, the kernel_size, stride, padding, and activation of Conv block-2 are set to 1, 1, “same,” and “relu,” respectively. It is worth noting that the WRI module does not use any DS or US modules to resize features, therefore, which can retain more LMF. Difference from the channelwise multiscale feature extracted by the rich-scale block, the feature extracted by the WRC module is a featurewise multiscale feature since the input of each branch is the same.

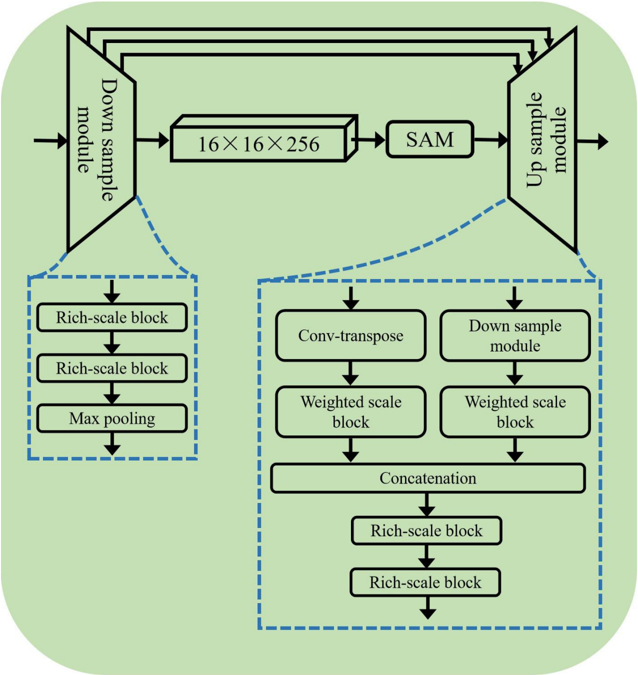
Down sample module
16×16×256
SAM
Up sample module
Rich-scale block
Rich-scale block
Max pooling
Conv-transpose
Down sample module
Weighted scale block
Weighted scale block
Concatenation
Rich-scale block
Rich-scale block

Fig. 5. WRC module.

E. WRC Module

As shown in Fig. 5, we design the WRC module, which is based on UNet and applies the weighted scale block and the rich-scale block to the DS module and the US module. The DS module consists of four DS operations. Each DS operation consists of two rich-scale blocks and one max-pooling operation. After each DS operation, the channels of the features are 32, 64, 128, and 256. The dimension ($W \times H \times C$) of the feature changed from $256 \times 256 \times 6$ to $16 \times 16 \times 256$ and then via SAM [25], [27] to obtain long-range spatial–temporal attention. The US module consists of four US operations. In each US operation, Conv-transpose is first applied to the feature. Before concatenate makes the skip connection with the feature of the same dimension in the DS module, the weighted scale block assigns appropriate weights to both. Two rich-scale blocks extract the multiscale feature. After each US operation, the channels of the features are 128, 64, 32, and 6.

It is worth noting that the WRC module will resize the feature during extracting the multiscale feature, which can extract more low-frequency features (semantic information) and filter noises.

III. EXPERIMENTS

A. Dataset

As shown in Fig. 6, to evaluate the performance of our proposed [weighted rich-scale inception coder network (WRICNet)] and state-of-the-art methods, we experiment on

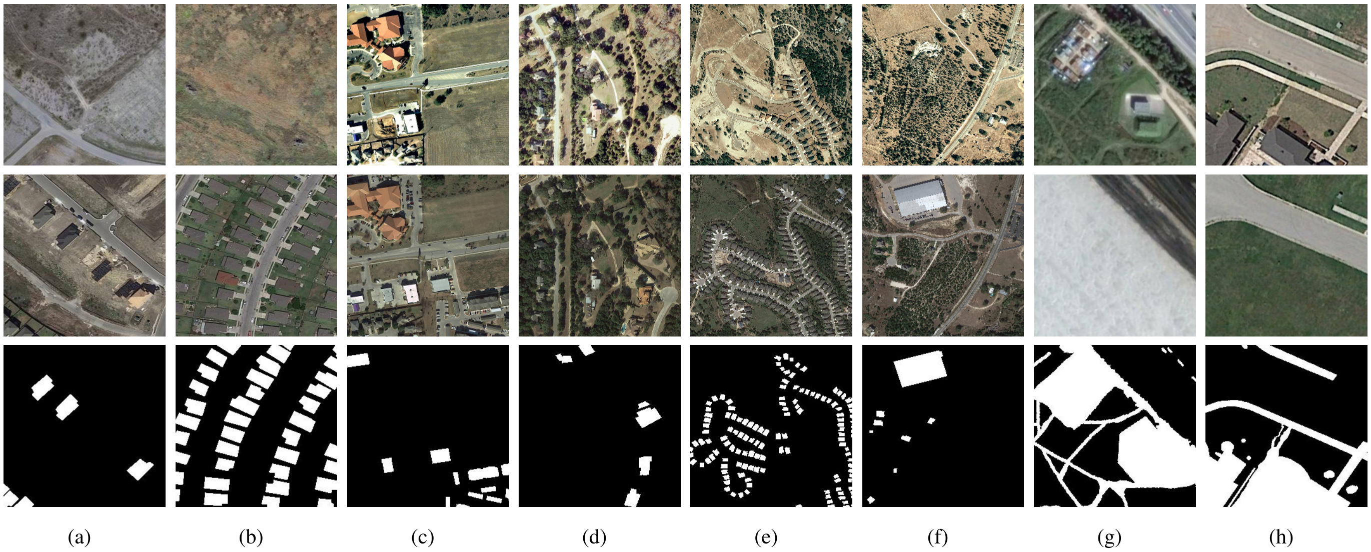

Fig. 6. Samples of datasets. (From Top to Bottom) Image T_1, image T_2, and ground truth. (a) and (b) Samples of LEVIR_CD HR dataset. (c) and (d) Samples of LEVIR_CD MR dataset. (e) and (f) Samples of LEVIR_CD LR dataset. (g) and (h) Samples of CDD HR dataset.

the LEVIR_CD and CDD high-resolution (HR) datasets; furthermore, to discuss the robustness of the model in different resolution dataset, we use bicubic interposition and nearest neighbor interpolation to resize image T_1, image T_2, and ground truth of the HR LEVIR_CD dataset. The HR dataset (1024 × 1024) reduces the resolution by two times to obtain the medium-resolution (MR) dataset (512 × 512) and reduces the resolution by four times to obtain the low-resolution (LR) dataset (256 × 256).

1) LEVIR_CD: The LEVIR_CD [25] HR dataset was collected from Google Earth and contains 637 pairs of HR images with a size of 1024 × 1024 pixels and 0.5 m/pixel of American cities in 2002 and 2018. Each pair of HR images consists of three images (image T_1, image T_2, and ground truth). As shown in Fig. 6(a) and (b), major changing areas of the LEVIR_CD HR dataset are buildings. Therefore, the changing area edge is clear and regular (e.g., square, rectangle, and circle).

2) CDD: The CDD [32] is a seasonal HR dataset with 0.03–1 m/pixel, which includes seven pairs of HR images with a size of 4725 × 2200 pixels and four pairs of images with a size of 1900 × 1000 pixels. After clipping and rotation, there are 10 000 training sets and 3000 test and validation sets. As shown in Fig. 6(g) and (h), major changing areas of the CDD HR dataset are buildings, roads, trees, and cars. Therefore, the changing area edge is unclear and irregular.

Since the size of the network input is 256 × 256, we crop the HR and MR dataset by a sliding window with a size of 256 × 256 and a stride of 256; to improve the robustness of the model, we randomly select a data enhancement method. Among them, the data enhancement methods include up-down flip, left-right flip, and 90° counterclockwise rotation. To speed up the convergence of the model during the training process, we use the min-max normalization to normalize the dataset, and the range is [0, 1].

B. Implementation Details

1) Experimental Settings: To improve the positive effect of features on backpropagation during training, we calculate the loss for features of different levels and construct a joint loss Loss. The Loss is calculated as follows:

$$\begin{aligned}\text{Loss} &= \lambda_1 L_{\text{ri}} + \lambda_2 L_{\text{wri}} + \lambda_3 L_{\text{ed}} \\ &\quad + \lambda_4 L_{\text{wed}} + \lambda_5 L_{\text{fu}} \end{aligned} \tag{1}$$

where λ_1, λ_2, λ_3, λ_4, and λ_5 are the wights of L_{ri}, L_{wri}, L_{ed}, L_{wed}, and L_{fu}, respectively. Based on experience and the parameter of the block or module, we set them to 1, 1.3, 0.5, 0.65, and 2.3. The cross-entropy losses L_{ri}, L_{wri}, L_{ed}, L_{wed}, and L_{fu} are calculated as follows:

$$L_{\text{ri}} = -\sum_{i=1}^{C} I_i^{\text{gt}} \times \log\left(I_i^{\text{ri}}\right) \tag{2}$$

$$L_{\text{wri}} = -\sum_{i=1}^{C} I_i^{\text{gt}} \times \log\left(I_i^{\text{wri}}\right) \tag{3}$$

$$L_{\text{ed}} = -\sum_{i=1}^{C} I_i^{\text{gt}} \times \log\left(I_i^{\text{rc}}\right) \tag{4}$$

$$L_{\text{wed}} = -\sum_{i=1}^{C} I_i^{\text{gt}} \times \log\left(I_i^{\text{wed}}\right) \tag{5}$$

$$L_{\text{fu}} = -\sum_{i=1}^{C} I_i^{\text{gt}} \times \log\left(I_i^{\text{fu}}\right) \tag{6}$$

where C is the category, which is set to 2, I^{gt} is the ground truth, I^{ri} is the output of the WRI module, and I^{wri} is the weighted map obtained from I^{ri} via the weighted scale block. I^{rc} is the output of the WRC module, and I^{wed} is the weighted map obtained from I^{rc} via the weighted scale block. I^{fu} is the CD output obtained by fusing the LMF and HMF. L_{ri} is the cross-entropy loss between I^{ri} and I^{gt}. L_{wri} is the cross-entropy loss between I^{wri} and I^{gt}. L_{ed} is the cross-entropy loss between I^{rc} and I^{gt}. L_{wed} is the cross-entropy loss between I^{wed} and I^{gt}. L_{fu} is the cross-entropy loss between I^{fu} and I^{gt}.

Our method is implemented based on the TensorFlow framework, and a single NVIDIA RTX 2080TI GPU with 11G

TABLE I

CHANGING AND UNCHANGING SAMPLES OF DATASETS

Resolution	Dataset	Changing	Unchanging
LR	LEVIR_CD	1337628	27825892
MR	LEVIR_CD	5349311	111304769
HR	LEVIR_CD	21412971	445203349
HR	CDD	80210466	575149534

memory is used for training and testing. The epoch and batch size are set to 200 and 1, respectively. We use Adam as an optimizer, and the initial learning rate, β_1, and β_2 are set to 1e^{-4}, 0.9, and 0.999, respectively.

Algorithm 1 Calculate Sample Weight of the Dataset

```
Input: $dateset, height, width$
Output: $c\_weight, uc\_weight$
function SampleWeight($dataset, height, width$)
    $count \leftarrow COUNT(dataset)$
    $c\_sample \leftarrow 0, uc\_sample \leftarrow 0$
    $img \leftarrow 0, i \leftarrow 0, j \leftarrow 0$
    while $img < count$ do
        while $i < height$ do
            while $j < width$ do
                if $dataset[img]_{i,j} = 1.$ then
                    $c\_sample \leftarrow c\_sample + 1$
                end if
                if $dataset[img]_{i,j} = 0.$ then
                    $uc\_sample \leftarrow uc\_sample + 1$
                end if
                $j \leftarrow j + 1$
            end while
            $i \leftarrow i + 1$
        end while
        $img \leftarrow img + 1$
    end while
    $c\_weight \leftarrow 1., c\_weight \leftarrow uc\_sample/c\_sample.$
    return $c\_weight, uc\_weight$
end function
```

According to Fig. 6 and Table I, we can find that the quantities of changing samples and unchanging samples in datasets differ greatly (the ratio of changing samples to unchanging samples in LR, MR, and HR LEVIR_CD is 1: 20.7, that in HR CDD is 1: 7.2). As we know, unbalanced samples in the dataset make it difficult for the model to fit well. Therefore, it is necessary to balance the number of changed and unchanged samples. As illustrated in Algorithm 1, we count the number of changed and unchanged samples during the training process and assign appropriate weights to the changed and unchanged samples.

In Algorithm 1, c_sample and uc_sample are the number of changed samples and unchanged samples, respectively. img$_{ij}$ is the value of the normalized at coordinates (i, j), c_weight is the weight of the changed samples, and uc_weight is the weight of the unchanged samples.

2) Comparative Method: To verify the effectiveness of the proposed WRICNet, some CD methods are compared and analyzed, which are as follows.

1) *FPN [14]:* A top-down architecture with skip connections leverages it as a feature pyramid, with predictions made independently at all levels.
2) *PANet [21]:* An end-to-end network based on FPN enhances the entire feature hierarchy with accurate localization signals in lower layers by bottom-up path augmentation.
3) *Res2Net [15]:* A building block for CNN by constructing hierarchical residual-like connections within one single residual block can represent multiscale features at a granular level and increase the range of RFs for each network layer.
4) *UNet++ [23]:* An improved UNet with 5 DS and US modules redesigns skip connections to aggregate features of varying semantic scales at the decoder subnetworks, leading to a highly flexible feature fusion scheme.
5) *FC-Siam-conc [22]:* A fully convolutional network with a siamese extension of the U-net model for CD combines a fully convolutional encoder–decoder paradigm and a siamese architecture that concatenates the skip connections during the decoding steps.
6) *STANet [25]:* A spatial–temporal attention neural network for RS image binary building CD uses SAM to capture the long-range spatial–temporal attention and extract the multiscale attention features, which is more robust to color and scale variations in bitemporal images.
7) *DASNet [24]:* An end-to-end network uses the SAM and CAM to capture long-range dependencies and obtain more discriminant feature representations to enhance recognition performance.
8) *EGRCNN [29]:* An end-to-end method for building CD, which incorporates both discriminative information and edge structure prior to improve the CD performance.

3) Evaluation Metrics: MA, FA, F1-Score (F1), and intersection over union (IoU) are used to quantitatively evaluate the performance of the comparative methods and the proposed method. These metrics are calculated as follows:

$$\mathrm{MA} = 1 - \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{7}$$

$$\mathrm{FA} = 1 - \frac{\mathrm{FP}}{\mathrm{TP} + \mathrm{FP}} \tag{8}$$

$$\begin{cases} P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ F1 = \frac{2PR}{P+R} \end{cases} \tag{9}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{10}$$

where false positive (FP) is the number of unpredicted unchanged pixels, true positive (TP) is the number of unchanged pixels detected correctly, false negative (FN) is the number of unpredicted changed pixels, and true negative (TN) is the number of changed pixels detected correctly. Among them, the F1 and IoU can better reflect the generalization ability of a model.

MA can intuitively reflect the probability of the true change in all pixels but detect no change. FA can intuitively reflect the probability of the true unchanged pixels but detected change. The lower the probability, the better the CD effect. F1 is the harmonic mean of Precision and Recall, and the IoU is the overlap between the predicted changed pixels and the changed pixels divided by the union area between them. F1 and IoU can comprehensively reflect the CD effect, and the higher the probability, the better the CD effect.

To evaluate the CD effect, we calculate the local optimal index (LOI) and the global index (GI), LOI reflects the optimal CD effect, and GI reflects the overall CD effect. First, we calculate the confusion matrix of each CD result and calculate the F1 of each CD result according to the formula. Second, sort each CD result according to F1 (from largest to smallest) and take the top 5%, 10%, and 100% of the CD results. Finally, calculate the confusion matrix of the selected CD result and calculate the MA, FA, F1, IoU of 5% LOI, 10% LOI, and GI according to (7)–(10).

C. Performance Experiment

1) Performance on the HR Datasets: The CD output of the comparative methods and the proposed method on HR datasets are shown in Fig. 7. Through careful observation, we can find that compared to FPN [Fig. 7(a)], PANet [Fig. 7(b)] strengthens the fusion of HR features and LR features via path aggregation, reducing MAs and FAs. Since UNet++ [Fig. 7(d)] uses multiple DS and US modules, there are few FAs outside the changing area, but there are abundant MAs in the changing area, and the changing area edge is not accurate enough. Compared with the previous three methods that can extract multiscale features, Res2Net [Fig. 7(c)] has a stronger ability to extract multiscale features, further reducing FAs outside the changing area and MAs in the changing area. The results of FC-Siam-conc [Fig. 7(e)], STANet [Fig. 7(f)], and DASNet [Fig. 7(g)] are relatively similar (there are a lot of MAs in the changing area), but the changing area edge of FC-Siam-conc [Fig. 7(e)] is more accurate, and there are more FAs at the changing area edge of STANet [Fig. 7(f)]. The EGRCNN [Fig. 7(h)] with additional edge information a prior has few MAs in the changing area, but the changing area edge is not accurate enough (there are a lot of FAs). The reason may be that the edges obtained by the Canny operator are not accurate enough. Compared with the comparison methods [Fig. 7(a)–(h)], our proposed WRICNet [Fig. 7(i)] has a stronger ability to extract multiscale features and complements LMF and HMF through adaptive weight fusion. This can further reduce FAs outside the changing area and MAs in the changing area and extract the accurate changing area edge. In addition, only Res2Net [Fig. 7(c)] and WRICNet [Fig. 7(i)], which have a stronger ability to extract multiscale features, can simultaneously detect the changing area with large differences in size. Overall, the CD output of the proposed method is the closest to the ground truth.

To quantitatively analyze the CD effect, we summarize the top 5%, 10% LOI, and GI of HR LEVIR_CD and CDD datasets in Tables II and III, respectively.

Through careful observation in Tables II and III, we can find that the performance of PANet is better than that of FPN. The MA of UNet++ is larger than others. The performance of FC-Siam-conc, STANet, and DASNet are relatively similar. The above conclusion is consistent with the result in Fig. 7.

In the 5% and 10% LOI of HR LEVIR_CD, WRICNet achieved the best FA, F1, and IoU. In GI, EGRCNN is slightly better than WRICNet, but the parameter of WRICNet is only 18.5% of EGRCNN. In the 5% and 10% LOI of HR CDD, WRICNet achieved the best MA, FA, F1, and IoU. In GI, Res2Net achieved the best MA, and WRICNet achieved the best FA, F1, and IoU. The performance of EGRCNN is quite different between HR LEVIR_CD and CDD because the changing area edge of HR LEVIR_CD is clear and regular, but the changing area edge of HR CDD is unclear and irregular. Since WRICNet does not rely on the changing area edge prior, accurate edges can be obtained on both HR LEVIR_CD and CDD.

2) Performance on the MR and LR LEVIR_CD Datasets: To discuss the robustness of the model in different resolution datasets, we experiment on MR and LR LEVIR_CD. Through careful observation in Fig. 8, we can find that the performance of all methods that have a certain degradation with the resolution decreases. The reason is that the smaller changing area can not be retrained, which reduces the proportion of the changing area in the dataset, and the changing area edge becomes unclear. Since EGRCNN relies on the changing edge prior, a significant loss in performance with the resolution decreases.

As shown in Tables IV and V, with the decrease in resolution, F1 and IoU of FPN are reduced by 22.5% and 21.9%, respectively. F1 and IoU of PANet are reduced by 10.8% and 13.4%, respectively. F1 and IoU of Res2Net are reduced by 1.3% and 1.8%, respectively. F1 and IoU of UNet++ are reduced by 4.2% and 5.5%, respectively. F1 and IoU of FC-Siam-conc are reduced by 4.7% and 6.6%, respectively. F1 and IoU of STANet are reduced by 4.7% and 6.1%, respectively. F1 and IoU of DASNet are reduced by 2.5% and 3.4%, respectively. F1 and IoU of EGRCNN are reduced by 19.3% and 24.5%, respectively. F1 and IoU of WRICNet are reduced by 7.0% and 10.5%, respectively. Though the performance degradation of WRICNet is not minimal (that of Res2Net is minimal), in the 5%, 10% LOI, and GI of MR and LR LEVIR_CD, WRICNet achieved the best F1 and IoU.

3) Comprehensive Performance on the HR, MR, and LR Datasets: The comparison of the above results on MR and LR LEVIR_CD uses the cropped result, which only reflects the partial result. To further comprehensively analyze the full result of the comparison methods and the proposed on HR, MR, and LR datasets, as shown in Fig. 9, we restore the cropped CD result on the HR and MR LERIC_CD datasets to a complete image and compares it with the output on the LR LERIC_CD dataset.

Through careful observation, as the resolution of the dataset decreases, the MAs of FPN [Fig. 9(a)], PANet [Fig. 9(b)], UNet++ [Fig. 9(d)], and FC-Siam-conc [Fig. 9(e)] increase a lot. The FAs at the edge of PANet [Fig. 9(b)], STANet [Fig. 9(f)], and EGRCNN [Fig. 9(h)] increase a lot. Moreover,

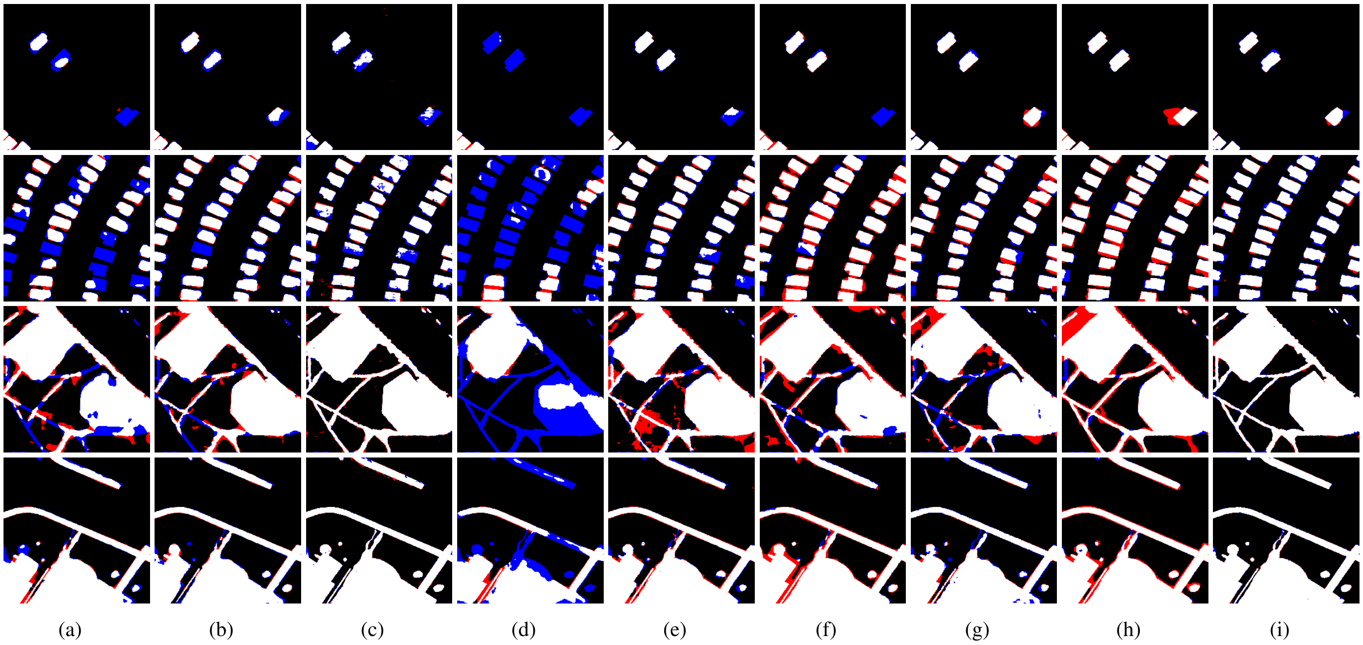

Fig. 7. Performance of the HR datasets. (a) FPN. (b) PANet. (c) Res2Net. (d) UNet++. (e) FC-Siam-conc. (f) STANet. (g) DASNet. (h) EGRCNN. (i) WRICNet. The top two rows are the results of the HR LEVIR_CD dataset, in which image T_1, image T_2, and ground truth are column of Fig. 6(a) and (b). The bottom two rows are the results of the HR CDD dataset, in which image T_1, image T_2, and ground truth are column of Fig. 6(g) and (h).

TABLE II

PERFORMANCE ON THE HR LEVIR_CD DATASET

Method	parameters	Local Optimal Index								Global Index			
		5%				10%							
		MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑
FPN	2,769,106	6.2	7.1	93.0	86.9	8.5	8.2	91.7	84.6	24.1	15.5	79.9	66.6
PANet	2,935,486	7.4	5.5	93.5	87.8	9.4	6.0	92.3	85.7	22.9	11.9	82.3	69.8
Res2Net	2,222,498	5.8	4.5	94.9	90.3	7.5	5.5	93.5	87.8	18.9	11.4	84.7	73.4
UNet++	9,043,074	6.9	6.4	93.3	87.5	8.4	8.0	91.8	84.8	27.3	16.2	77.8	63.7
FC-Siam-conc	3,442,954	4.4	5.3	95.1	90.7	5.7	6.1	94.1	88.9	13.4	10.7	87.9	78.4
STANet	16,929,152	3.5	7.4	94.5	89.6	3.8	8.9	93.6	87.9	**8.6**	15.0	88.1	78.7
DASNet	28,941,962	4.6	6.6	94.4	89.3	5.8	6.8	93.7	88.2	16.6	9.4	86.9	76.8
EGRCNN	15,726,554	**2.8**	7.0	95.1	90.6	**3.1**	8.4	94.2	89.0	9.0	13.2	**88.6**	**79.7**
WRICNet	2,902,783	4.3	**4.0**	**95.9**	**92.1**	5.9	**4.4**	**94.8**	**90.2**	16.1	**7.2**	88.3	79.0

TABLE III

PERFORMANCE ON THE HR CDD DATASET

Method	parameters	Local Optimal Index								Global Index			
		5%				10%							
		MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑
FPN	2,769,106	1.0	0.9	99.0	98.1	1.7	1.6	98.4	96.8	13.1	7.5	89.6	81.2
PANet	2,935,486	0.8	0.7	99.2	98.5	1.3	1.2	98.7	97.5	8.4	7.0	92.3	85.7
Res2Net	2,222,498	0.4	0.3	99.7	99.4	0.5	0.4	99.5	99.1	**2.4**	2.3	97.6	95.3
UNet++	9,043,074	6.8	2.0	95.5	91.5	10.6	3.3	92.9	86.7	39.2	19.0	69.5	53.2
FC-Siam-conc	3,442,954	0.9	0.9	99.1	98.3	1.4	1.5	98.6	97.2	10.9	7.4	90.9	83.2
STANet	16,929,152	0.8	1.8	98.7	97.4	1.6	3.0	97.7	95.5	11.5	8.4	89.1	80.5
DASNet	28,941,962	1.1	0.8	99.0	98.1	1.8	1.4	98.4	96.9	11.0	6.9	91.0	83.8
EGRCNN	15,726,554	**0.2**	1.3	99.3	98.5	0.3	2.3	98.6	97.3	3.6	12.0	92.0	85.2
WRICNet	2,902,783	**0.2**	**0.2**	**99.8**	**99.5**	**0.4**	**0.3**	**99.6**	**99.3**	2.8	**1.4**	**97.9**	**95.9**

when the changing area has a large difference in size, the MAs in the larger change area of FPN [Fig. 9(a)], PANet [Fig. 9(b)], UNet++ [Fig. 9(d)], and FC-Siam-conc [Fig. 9(e)] increase a lot. Although Res2Net [Fig. 9(c)] and DASNet [Fig. 9(g)] can achieve better results, the result of WRICNet is the closest to ground truth (less MAs and FAs and more accurate changing

Note: the sub-captions (a)–(i) are inside the figure box.

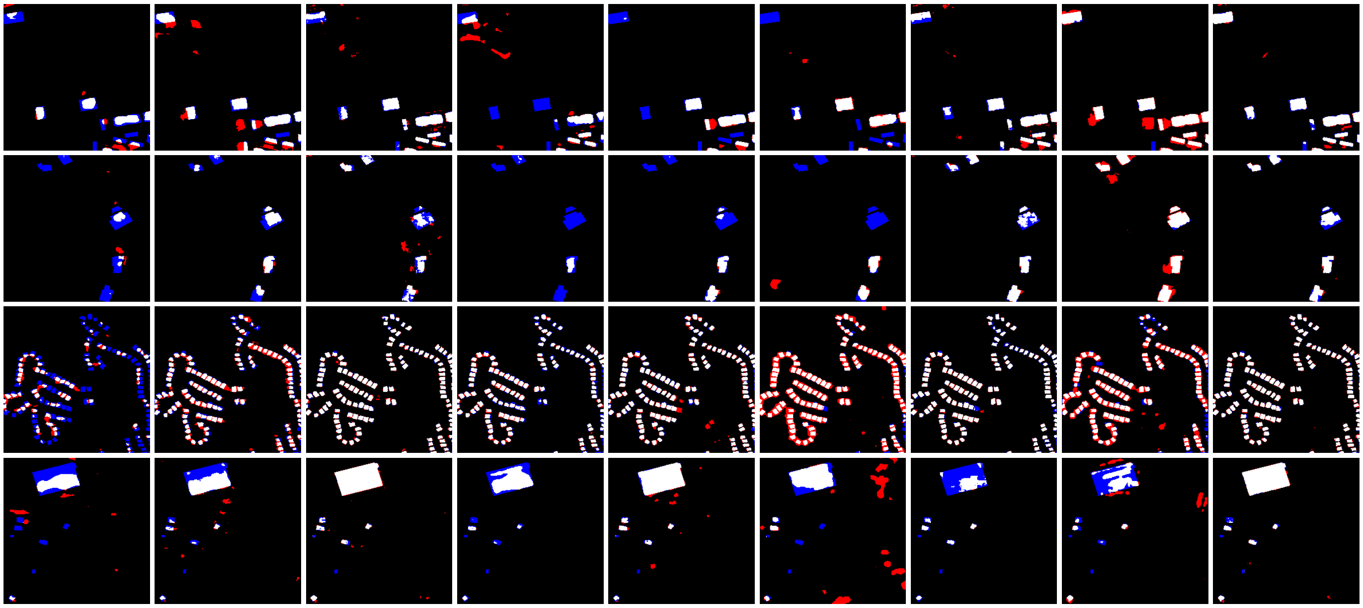

Fig. 8. Performance of the MR and LR LEVIR_CD datasets. (a) FPN. (b) PANet. (c) Res2Net. (d) UNet++. (e) FC-Siam-conc. (f) STANet. (g) DASNet. (h) EGRCNN. (i) WRICNet. The top two rows are the results of the MR LEVIR_CD dataset, in which image T_1, image T_2, and ground truth are column of Fig. 6(c) and (d). The bottom two rows are the results of the LR LEVIR_CD dataset, which image T_1, image T_2, and ground truth are column of Fig. 6(e) and (f).

TABLE IV
PERFORMANCE ON THE MR LEVIR_CD DATASET

Method	parameters	Local Optimal Index								Global Index			
		5%				10%							
		MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑
FPN	2,769,106	15.9	12.9	85.6	74.8	17.7	14.4	84.0	72.4	41.0	21.0	67.3	50.8
PANet	2,935,486	12.9	7.9	89.5	81.0	14.3	8.2	88.7	79.6	28.7	13.4	78.2	64.2
Res2Net	2,222,498	6.2	6.1	93.9	88.5	7.6	6.5	92.9	86.8	21.0	13.7	82.5	70.2
UNet++	9,043,074	15.1	9.1	87.8	78.2	18.0	10.3	85.7	75.0	35.8	15.1	72.7	57.6
FC-Siam-conc	3,442,954	6.9	6.7	93.2	87.2	8.2	7.4	92.2	85.5	18.6	13.4	83.9	72.2
STANet	16,929,152	7.8	12.8	89.6	81.2	8.1	14.7	88.5	79.3	27.4	17.3	77.3	63.0
DASNet	28,941,962	9.7	**4.4**	92.9	86.7	11.7	**4.7**	91.6	84.6	30.3	**5.8**	80.1	66.8
EGRCNN	15,726,554	**3.4**	12.7	91.7	84.7	**3.5**	14.9	90.4	82.5	**7.3**	24.0	83.5	71.7
WRICNet	2,902,783	5.3	5.6	**94.6**	**89.7**	6.5	5.9	**93.8**	**88.4**	10.7	9.6	**88.2**	**78.9**

TABLE V
PERFORMANCE ON THE LR LEVIR_CD DATASET

Method	parameters	Local Optimal Index								Global Index			
		5%				10%							
		MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑
FPN	2,769,106	29.4	20.8	74.7	59.6	34.8	25.8	69.4	53.2	65.4	36.6	44.8	28.9
PANet	2,935,486	21.3	15.4	81.6	68.9	19.8	19.7	80.2	67.0	37.4	27.1	67.4	50.8
Res2Net	2,222,498	8.2	9.2	91.3	83.9	8.6	10.4	90.5	82.6	21.8	15.5	**81.2**	**68.4**
UNet++	9,043,074	15.7	11.0	86.6	76.3	16.6	12.0	85.6	74.9	42.9	14.3	68.5	52.1
FC-Siam-conc	3,442,954	9.9	8.9	90.6	82.8	10.5	11.2	89.2	80.4	23.1	18.4	79.2	65.6
STANet	16,929,152	**6.5**	22.3	84.9	73.8	8.3	23.9	83.2	71.2	21.4	32.6	72.6	56.9
DASNet	28,941,962	11.2	**7.9**	90.4	82.5	11.1	**8.8**	90.0	81.8	31.4	**10.7**	77.6	63.4
EGRCNN	15,726,554	14.1	28.1	78.3	64.3	11.4	33.4	76.0	61.3	26.6	43.0	64.2	47.2
WRICNet	2,902,783	7.2	9.1	**91.8**	**84.9**	**7.8**	10.8	**90.6**	**82.9**	**22.0**	15.3	**81.2**	**68.4**

edge). The above proves that the proposed method has strong robustness to multiresolution datasets.

4) Comparison of the Output of WRI and WRC Module: To verify the effectiveness of the WRI module to extract LMF, the effectiveness of the WRC module to extract HMF, and the ability of the weighted scale block to assign appropriate weights, as shown in Fig. 10, we compare the I^{ri}, heatmap of I^{wri}, I^{rc}, and heatmap of I^{wrc} on the LR LEVIR_CD, MR LEVIR_CD, HR LEVIR_CD, and HR CDD dataset, respectively.

Through careful observation, we found some of the following rules.

1) In I^{ri} [Fig. 10(d)], there are fewer MAs in the changing area but more FAs outside the changing area.

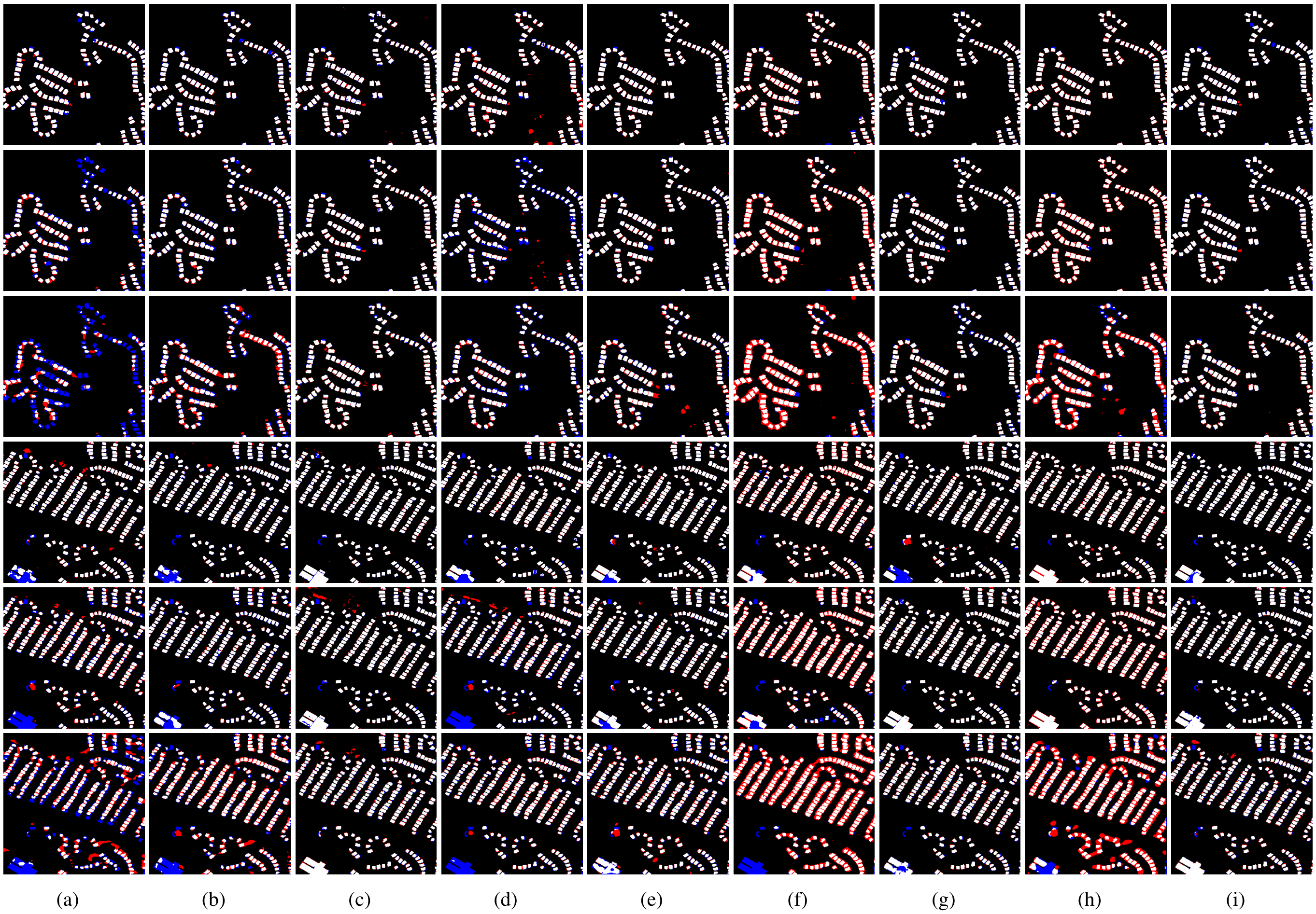

Fig. 9. Comprehensive performance of the HR, MR, and LR LEVIR_CD datasets. (a) FPN. (b) PANet. (c) Res2Net. (d) UNet++. (e) FC-Siam-conc. (f) STANet. (g) DASNet. (h) EGRCNN. (i) WRICNet. (From Top to Bottom) First and fourth rows are results of the HR LEVIR_CD dataset, second and fifth rows are results of the MR LEVIR_CD dataset, and third and sixth rows are results of the LR LEVIR_CD dataset.

2) In I^{rc} [Fig. 10(f)], there are more MAs in the changing area but fewer FAs outside the changing area.
3) In the heatmap of I^{wri} and I^{wrc}, the area with the greatest weight is mainly located at the changing area edge; moreover, the edge of I^{wri} is accurate, and the edge of I^{wrc} is inaccurate.
4) Compared to I^{ri} [Fig. 10(d)] and I^{rc} [Fig. 10(f)], I^{fu} has fewer MAs in the changing area and fewer FAs outside the changing area, and the changing area edge is more accurate. Overall, I^{fu} is closest to the ground truth.

According to the above, it can be proven that the WRI module and the WRC module can obtain LMF and HMF, respectively. WRICNet can complement LMF and HMF well via adaptive weight fusion.

5) Ablation Study: To verify the effectiveness of our network, we compare our WRICNet with its ablated versions:

1) *WRICNet:* Our proposed.
2) *w/o Multichannel:* Changing the network input structure of the proposed method from concatenation to Siamese+concatenation.
3) *w/o Weighted Scale Block:* Removing the weighted scale block of the WRI module and WRC module.
4) *w/o Rich-Scale Block:* Replacing the rich-scale block with a single Conv block-3.
5) *Rich-Scale Block-Scale2:* Changing scale of the rich-scale block to 2.
6) *Rich-Scale Block-Scale3:* Changing scale of the rich-scale block to 3.
7) *Rich-Scale Block-Scale5:* Changing scale of the rich-scale block to 5.
8) *w/o Inception Module v2:* Removing the improvement of the Inception module in this article.
9) *w/o Rich-Scale Block v2:* Removing the improvement of the rich-scale block in this article.
10) *w/o Weighted Class:* In the training process, the appropriate weights are no longer assigned to the changed and unchanged samples to solve the problem of unbalanced sample distribution in the dataset, that is, the weight of the changed and unchanged samples in the training process is set to 1:1.

As shown in Table VI, in the ablation study, the LOI and GI are average values on the HR, MR, and LR LEVIR_CD datasets.

Through careful observation, we found that some of the following rules are given.

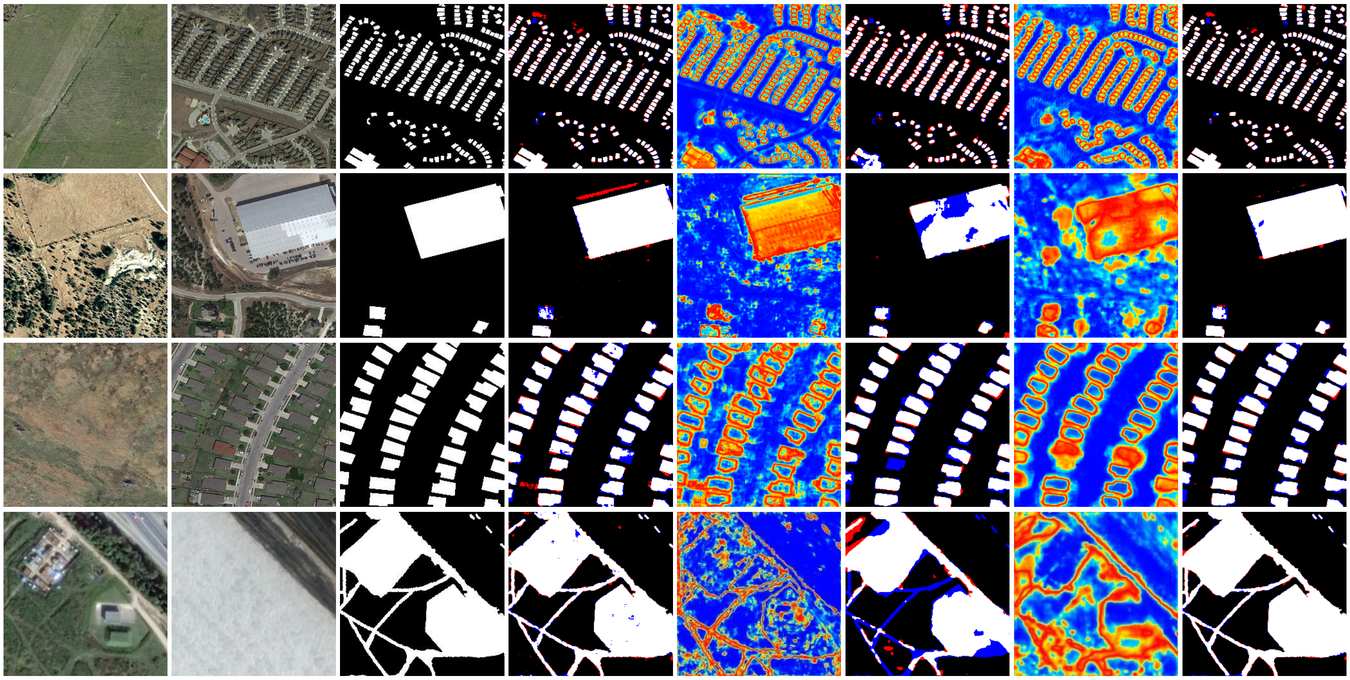

Fig. 10. Outputs of WRICNet on the HR, MR, and LR datasets. (a) Image T_1. (b) Image T_2. (c) Ground truth. (d) I^{ri}. (e) I^{wri}. (f) I^{rc}. (g) I^{wrc}. (h) I^{fu}. (From Top to Bottom) Results of the LR LEVIR_CD, MR LEVIR_CD, HR LEVIR_CD, and HR CDD dataset.

TABLE VI

ABLATION STUDY OF WRICNET ON LR-, MR, AND HR LEVIR_CD DATASETS

Method	parameters	Local Optimal Index								Global Index			
		5%				10%							
		MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑	MA↓	FA↓	F1↑	IoU↑
WRICNet	2,902,783	5.6	6.2	**94.1**	**88.9**	6.7	7.0	93.1	87.2	18.0	**11.1**	**85.4**	**74.6**
w/o multi-channel	2,896,932	6.1	6.6	93.6	88.0	7.4	6.9	92.9	86.7	17.4	12.0	85.2	74.2
w/o weighted scale block	2,591,503	6.4	5.8	93.7	88.5	7.3	6.5	92.9	87.1	18.3	**11.1**	84.9	73.9
w/o rich-scale block	3,706,491	7.7	7.9	92.3	85.7	8.8	8.5	91.3	84.2	20.6	13.4	82.8	70.8
rich-scale block-scale2	2,434,775	6.9	**5.4**	93.8	88.3	8.3	**5.9**	92.9	86.8	22.8	9.9	83.0	71.0
rich-scale block-scale3	3,191,947	7.6	5.9	93.2	87.4	9.0	6.5	92.2	85.6	21.8	12.4	82.6	70.4
rich-scale block-scale5	3,476,215	5.6	6.5	93.9	88.6	7.0	6.8	93.1	87.1	**17.2**	11.6	85.3	74.5
w/o inception module v2	3,834,239	5.9	6.8	93.6	88.0	7.2	7.3	92.7	86.5	19.4	12.3	83.8	72.4
w/o rich-scale block v2	2,202,811	**5.5**	6.6	93.9	88.6	**6.6**	7.0	**93.2**	**87.3**	18.3	11.7	84.5	73.5
w/o weighted class	2,902,783	6.5	6.2	93.7	88.1	8.1	6.7	92.6	86.2	19.6	11.9	84.0	72.4

1) WRICNet achieves best F1 (94.1%) and IoU (88.9%) of top 5% LOI and best FA (11.1%), F1 (85.4%), and IoU (74.6%) of GI.
2) Compared to w/o multichannel, WRICNet has slightly more parameters, whereas F1 increased by 0.5%, 0.2%, and 0.2% in the top 5%, 10% LOI, and GI, respectively, and IoU increased by 0.9%, 0.5%, and 0.4% in the top 5%, 10% LOI, and GI, respectively. The above demonstrates that the Siamese structure can slightly reduce the model parameters, but it fails to improve the CD effect in our work.
3) Compared to the w/o weighted scale block, F1 of the proposed has increased by 0.4%, 0.2%, and 0.5% in the top 5%, 10% LOI, and GI, respectively, and IoU increased by 0.4%, 0.1%, and 0.7% in the top 5%, 10% LOI, and GI, respectively. In our work, it was proven that the weighted scale block, which assigns appropriate weights to features of different scales, is effective.
4) Compared to w/o rich-scale block, F1 of the proposed increased by 1.8%, 1.8%, and 2.6% in the top 5%, 10% LOI, and GI, respectively, and IoU increased by 3.2%, 3.0%, and 3.8% in the top 5%, 10% LOI, and GI, respectively. The proposed parameters are 21% less than the parameters of the w/o rich-scale block. The above demonstrates that the rich-scale block can reduce model parameters and improve the CD effect by extracting multiscale features.
5) Compared to rich-scale block-scale2, rich-scale block-scale3, and rich-scale block-scale5, WRICNet achieves better F1 and IoU of top 5%, 10% LOI, and GI.

The above demonstrates that the optimal scale of the rich-scale block is 4 (setting of WRICNet).

6) Compared to the w/o inception module v2, F1 of the proposed method increased by 0.5%, 0.4%, and 1.6% in the top 5%, 10% LOI, and GI, respectively, and IoU increased by 0.9%, 0.7%, and 2.2% in the top 5%, 10% LOI, and GI, respectively. The proposed parameters are 24% less than the parameters of w/o inception module v2. Since reducing the model parameters, which speeds up the convergence of the model, the CD effect improved. Overall, it is effective for the improvement of the inception module.
7) Compared to the w/o rich-scale block v2, F1 of the proposed method increased by 0.2% and 0.9% in the top 5% LOI, and GI, respectively, and IoU increased by 0.3% and 1.1% in the top 5% LOI, and GI, respectively, which proves that the improvement of the rich-scale block in this article is effective. Although the number of model parameters increased by 32%, the effect of the CD improved.
8) Compared to w/o weighted class, F1 of the proposed method increased by 0.4%, 0.5%, and 1.4% in the top 5%, 10% LOI, and GI, respectively, and IoU increased by 0.8%, 1.0%, and 2.2% in the top 5%, 10% LOI, and GI, respectively. It is proven that the training strategy of assigning appropriate weights to the changed samples and the unchanged samples in the training process to achieve sample balance is effective.

IV. LIMITATION

We use two branches to extract LMF and HMF. In addition, we use many branches and grouping structures to extract multiscale features, further reducing FAs outside the changing area and MAs in the changing area. Meanwhile, it will increase network calculations (training and testing speed is slow) and GPU memory consumption (the batch size only can be set to 1 on the 2080TI with 11-GB memory).

V. CONCLUSION

To extract both low-level features and high-level features at the same time, we directly use the two branches to extract them separately. To make the model robust to multiresolution datasets and simultaneously detect the changing area with large differences in size, we use an inception module and an UNet to extract multiscale features in the two branches. Moreover, we use the rich-scale module to strengthen the extraction of multiscale features. Furthermore, we improve the inception and rich-scale module for reducing parameters and strengthening channelwise multiscale features extraction. To complement LMF and HMF, we design a weighted scale model that assigns appropriate weights to multiscale features. According to the above, we design WRICNet.

The performance experiment on the HR, MR, and LR datasets prove that the proposed WRICNet model can complement LMF and HMF well, therefore reducing the MAs in the changing area and FAs outside the changing area and making the changing area edge more accurate. Therefore, the CD outputs of the HR, MR, and LR datasets are closest to the ground truth. Furthermore, the LOI, GI, and model parameters of the comparative methods and the proposed method prove that the proposed method can achieve a better CD effect based on fewer model parameters. The I^{ri}, heatmap of I^{wri}, I^{rc}, and heatmap of I^{wrc} on the HR, MR, and LR datasets prove that the WRI module and the WRC module can obtain LMF and HMF, respectively, and WRICNet can complement LMF and HMF well. Finally, the ablation study illustrates that the training strategy (w/o weighted class), the model settings (w/o multichannel, scale of rich-scale block), the proposed weighted scale block, and the improvements of the rich-scale block and inception module are effective.

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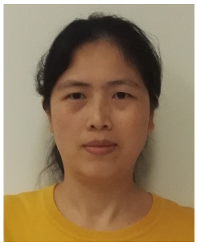

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